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Operation

Sequence of Operation

Your new **BluePrint** Automation **Collator 200** system is made up of these primary areas.

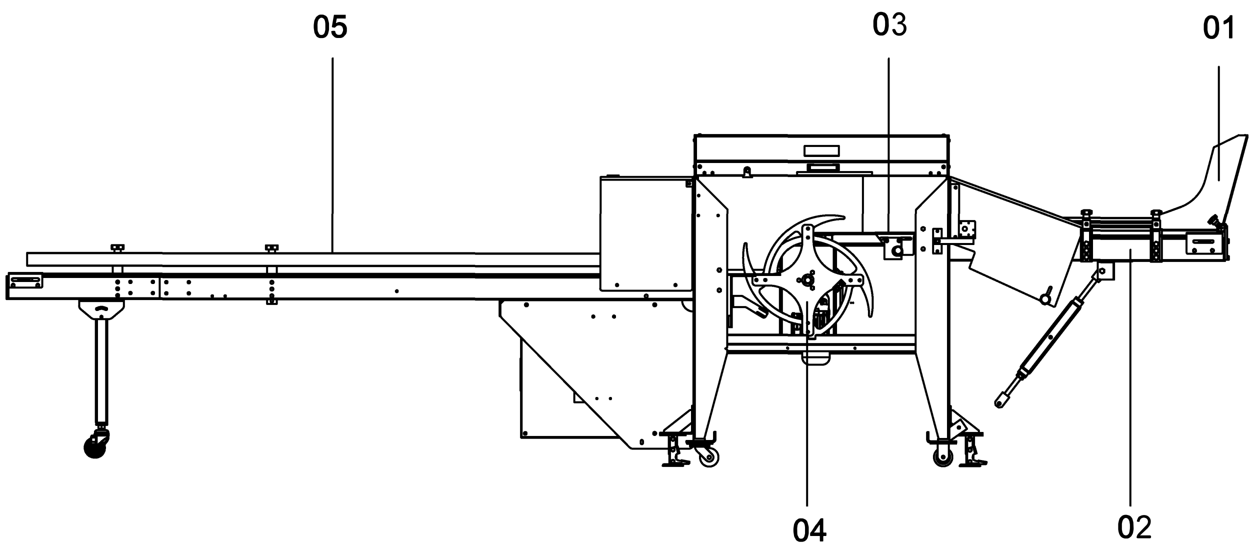
Pos 01: Product In-feed Chute

Pos 02: Product In-feed Conveyor (Feed belt conveyor with “extra grip”)

Pos 03: Count Conveyor

Pos 04: Collator Wheel

Pos 05: Exit Conveyor (Discharge Conveyor)

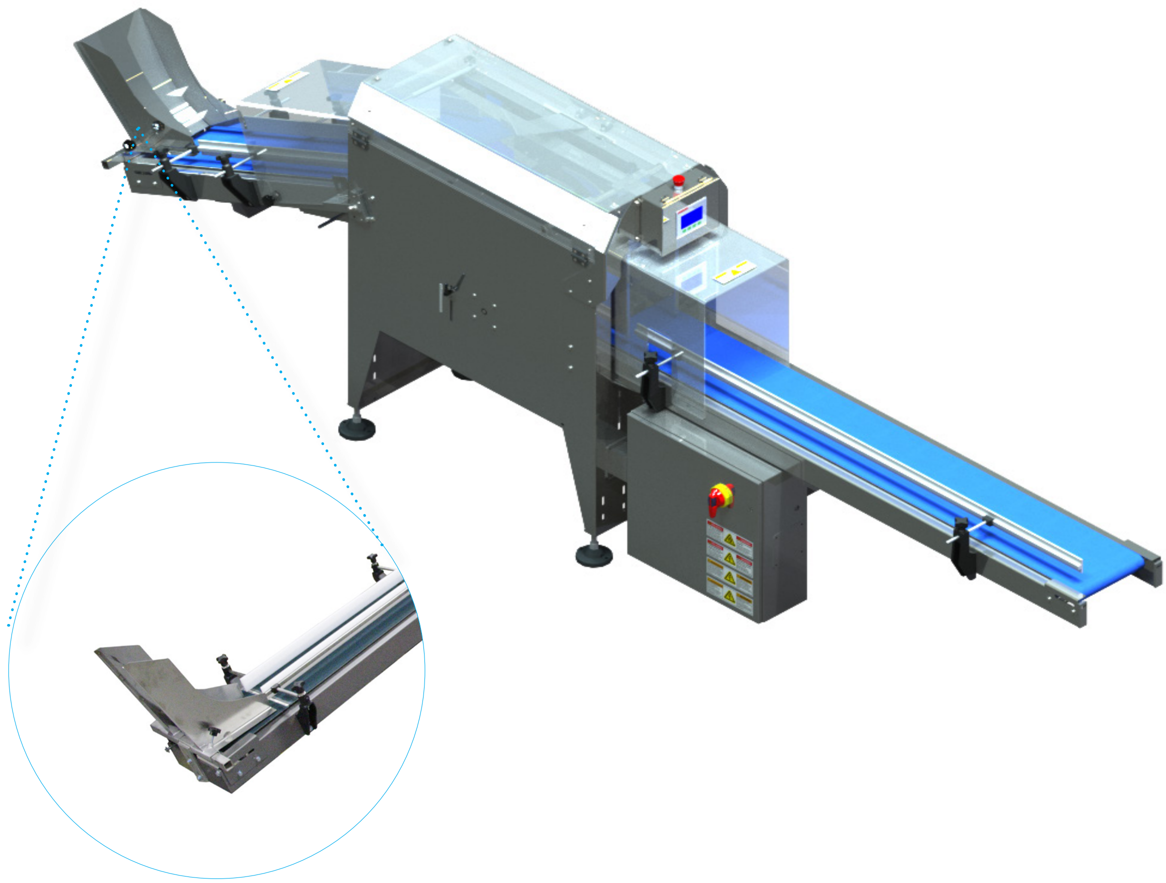


Packages are dropped from upstream equipment into the product in-feed chute which aligns the packages to the centre of the product in-feed conveyor belt. The product in-feed conveyor transports the packages to the count belt. The count belt moves the packages past the count sensor to the collator wheel. The collator wheel rotates and transfers the packages to the exit conveyor in a single row arranged vertically front to back. From the exit conveyor, the packages can be hand packed into cases.

Product In-feed Chute

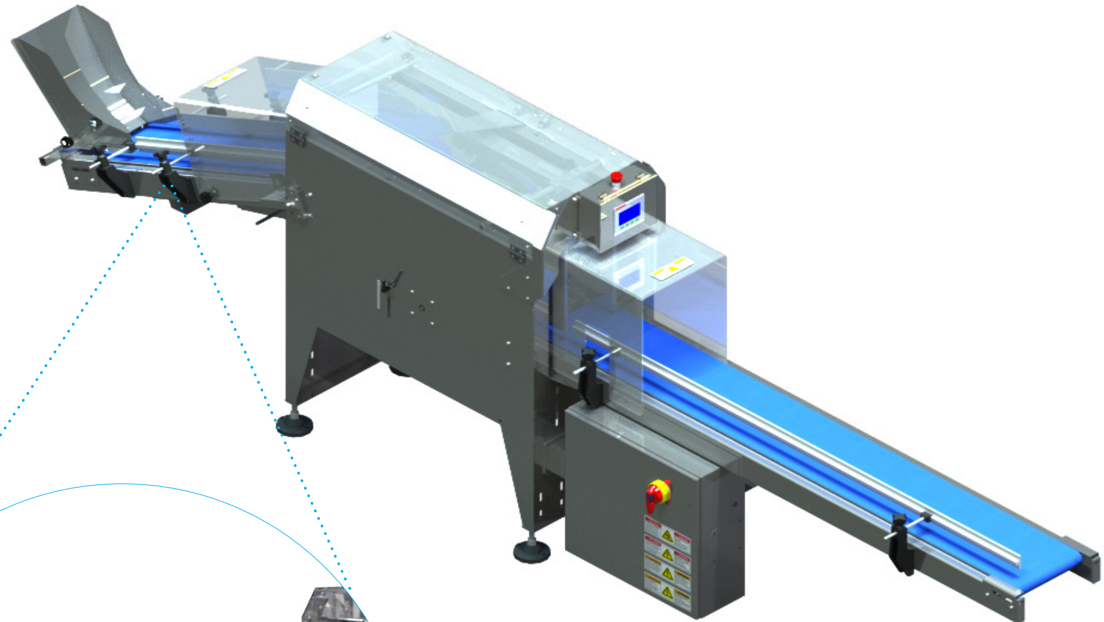
01

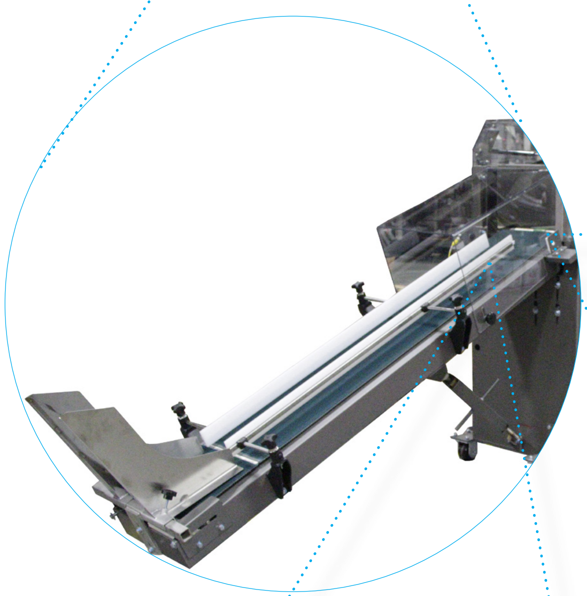
The packages are dropped from the upstream equipment onto the product in-feed conveyor via the product in-feed chute. The chute is located on the lower end of the product in-feed conveyor.

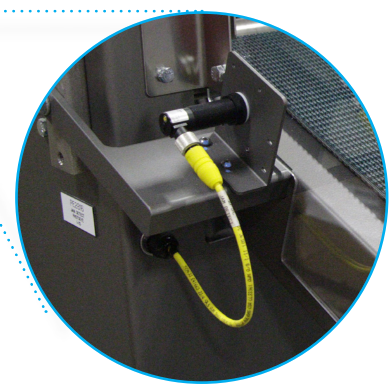


Product In-feed Conveyor

The packages dropped from the product in-feed chute are conducted by the in-feed conveyor into the machine.

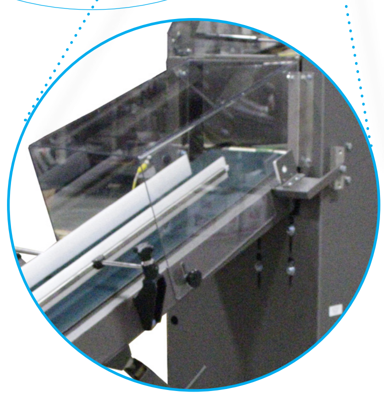






Jam Detection Photoelectric Sensor

The jam detection photoelectric sensor is located toward the end of the in-feed conveyor belt. This sensor will signal a product jam by detecting product presence for a duration of time exceeding the predefined set point.

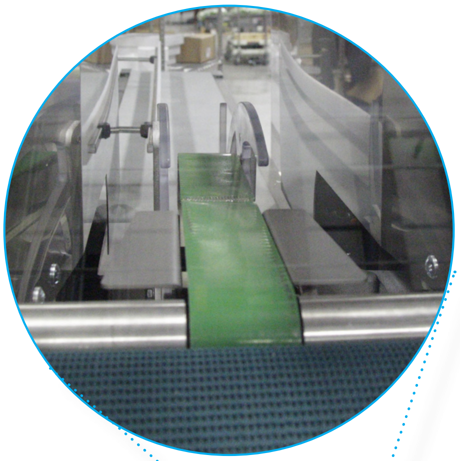


In-feed Tunnel Guarding

This guard is provided to protect the operator when the machine is running by limiting access via the in-feed opening.

Count Belt

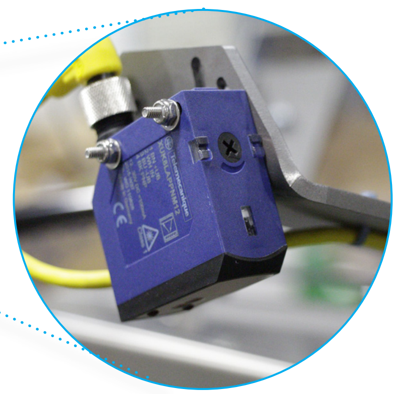
The count belt receives the packages from the product in-feed conveyor and moves them past the count sensor and into the collator wheel.

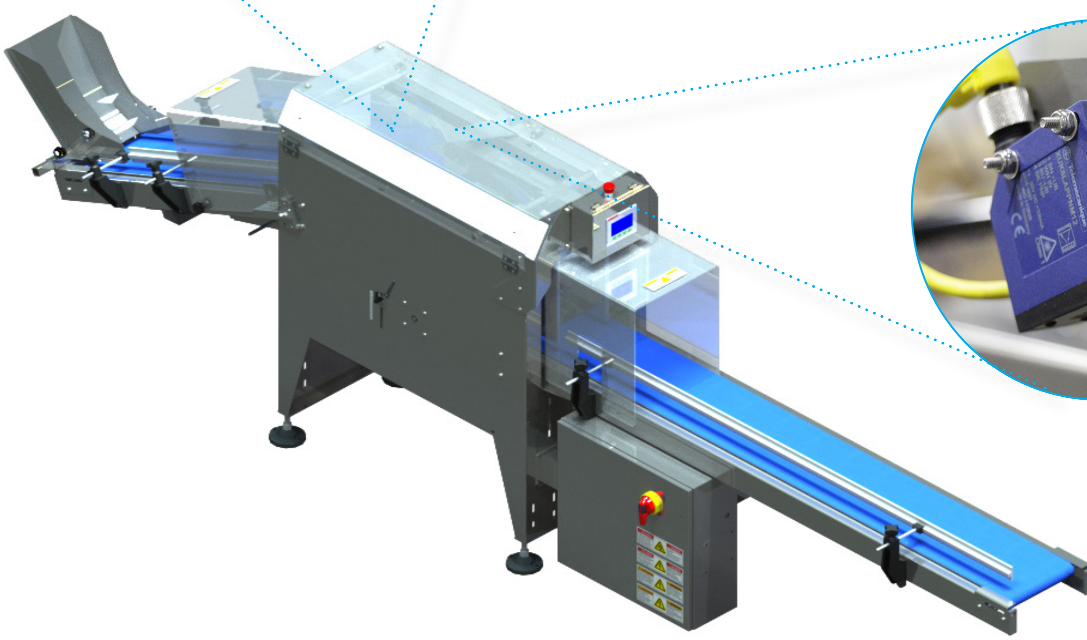


CLASS 1 LASER PRODUCT

Package Count Photoelectric Sensor

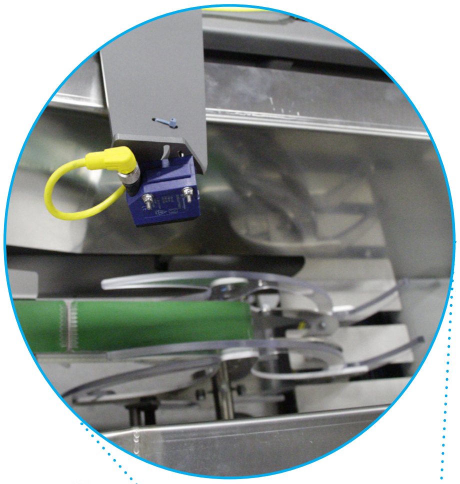
This diffuse laser photoelectric sensor detects the package presence, logs a count in the shift output field of the production data screen on the HMI, and signals the collator wheel to index. This sensor is an **IEC 60825-1 Class 1 Laser product**.

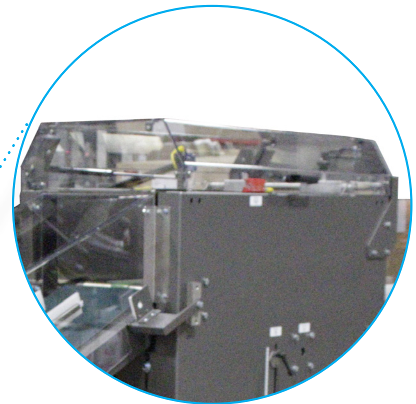


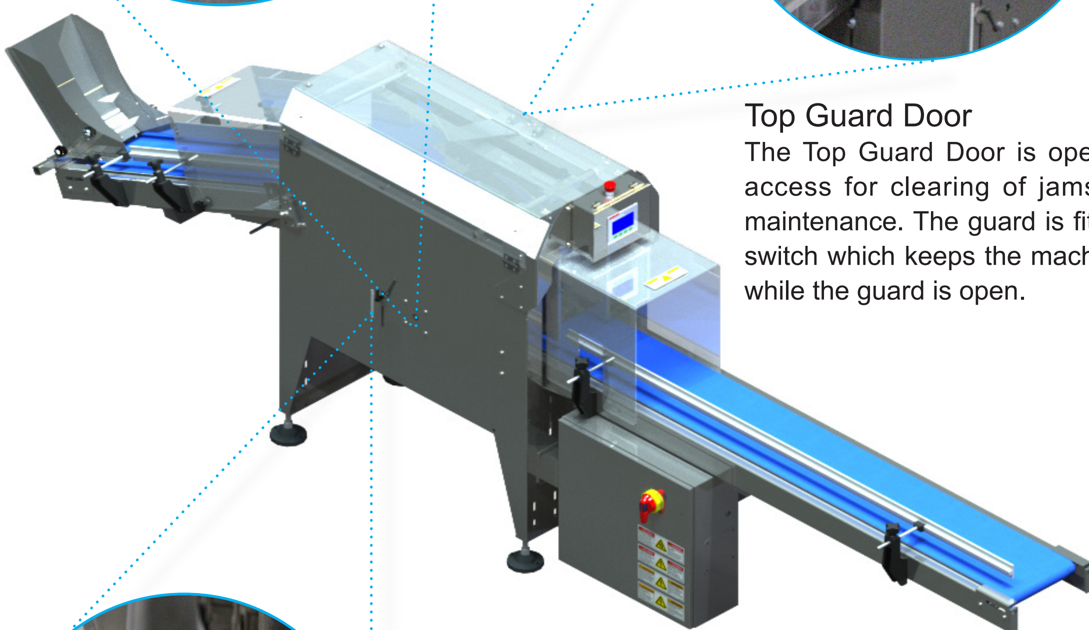


Collator Wheel

The collator wheel collects one product package at a time. As the package moves from the count belt to the collator wheel, the wheel rotates and delivers the package stood upright to the exit conveyor.







Top Guard Door

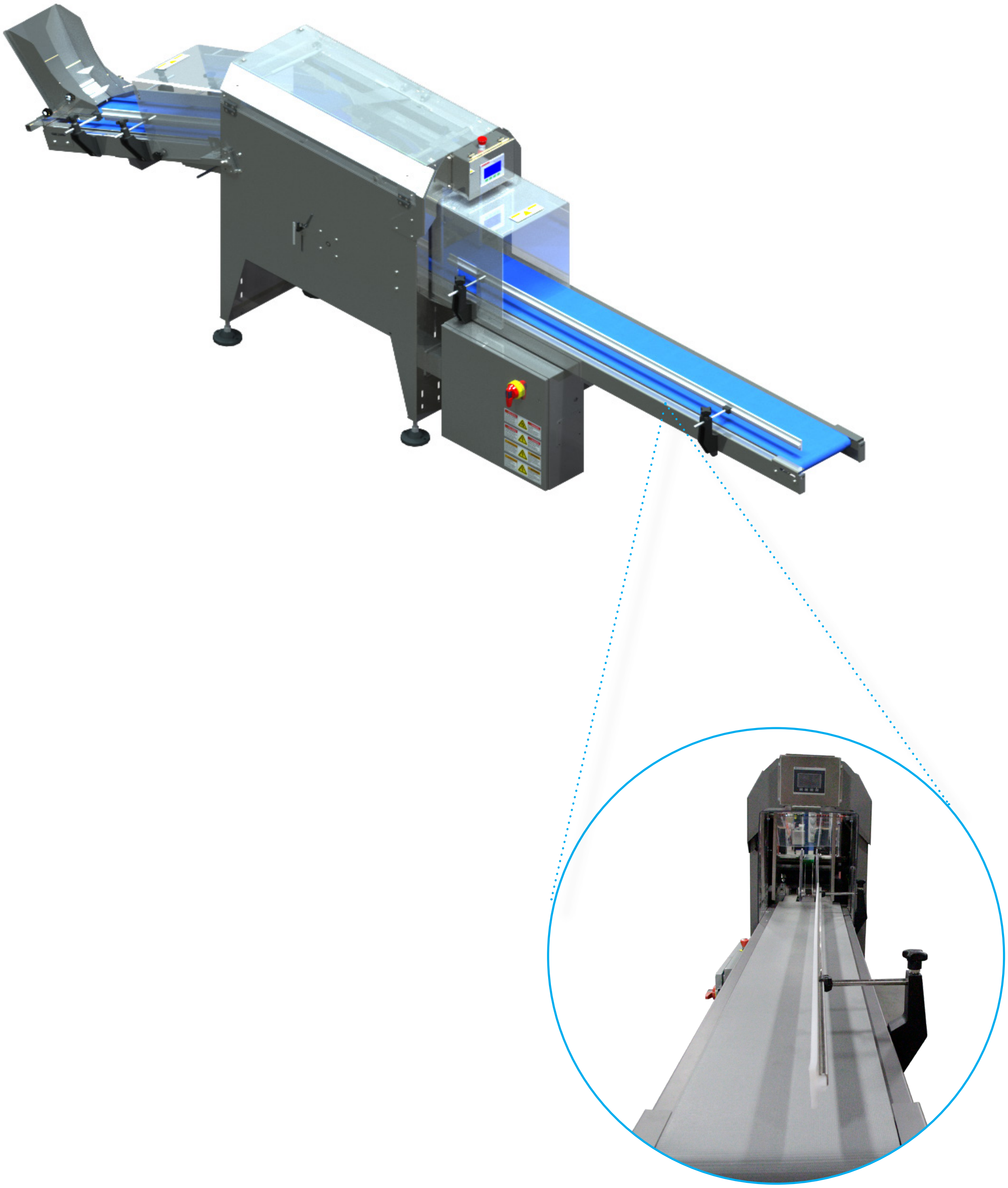
The Top Guard Door is openable; providing access for clearing of jams, cleaning, and maintenance. The guard is fitted with a safety switch which keeps the machine from running while the guard is open.

Wheel Stop Photoelectric Sensor

This polarized retro-reflective sensor detects the position of the wheel as it rotates to signal the end of rotation.

Exit Conveyor

The exit conveyor receives the package from the Collator Wheel then moves (indexes) each time a package is received. The bags can then be hand packed into cases. Any bags that did not get packed will fall off the exit conveyor into a collection bin.



Operator Controls

The operator's position must allow for continuous checking of the product flow and viewing of the human/machine interface [**HMI**]; hence, the normal operator position is defined as approximately 1 meter from the HMI. The operator controls are detailed below:

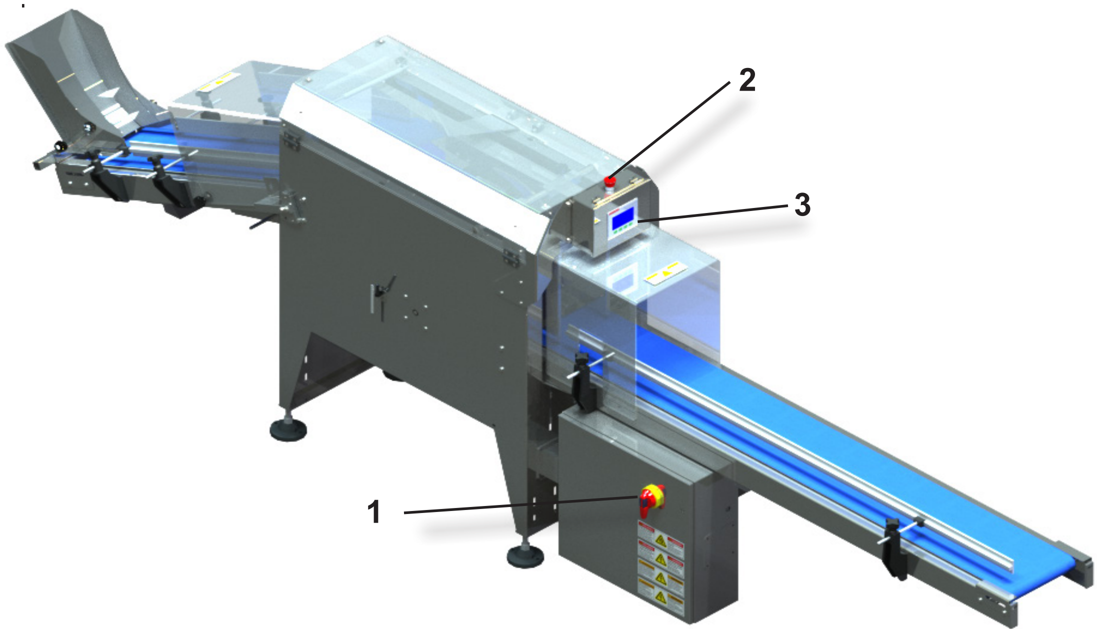
Electrical Disconnect Switch

The main power is disconnected at the main electrical disconnect switch (1) located on the main electrical cabinet. The switch should be disengaged to protect against general electrical danger from line surges, thunderstorms, or during machine maintenance or repair tasks that do not require that the machine be power

Human/Machine Interface

The machine is furnished with an HMI (3) interactive video display monitor that displays all the machine status and diagnostic information. The HMI (3) allows the operator to be informed in real-time of the status of the machine.

Each of the HMI screens are depicted and described in the following paragraphs.



Emergency Stop Push-button

To stop the machine in an emergency situation (i.e., imminent danger, mechanical accident, etc.), press the emergency stop push-button (2) located on the HMI station (3). When depressed, the emergency stop push-button (2) provides deactivation of all power to all moving mechanisms, motor controls, and PLC output modules. All devices that do not provide or enable motion will remain in a powered state, providing fault indication to the operator. To enable the machine to restart, the emergency stop push-button must be manually twisted to release.

NOTICE

To avoid unnecessary stress on mechanical parts, use the emergency stop push-button exclusively in the event of a REAL EMERGENCY. For routine shut-down, use the Service Stop push-button on the HMI screen.

Pre Start-up Procedures

⚠ WARNING

Articles of jewellery and accessories such as watch-bands, bracelets, rings, key chains, necklaces, loose fitting clothing and neck-ties are NOT to be worn while operating the machinery. Make sure that all personnel in the production environment are wearing the requisite personal protection equipment [PPE] such as hearing protection, safety glasses, and non-slip safety shoes.

At the start of a production cycle, the machine operator should check and verify the following conditions obtain:

• Compliance with all safety regulations.

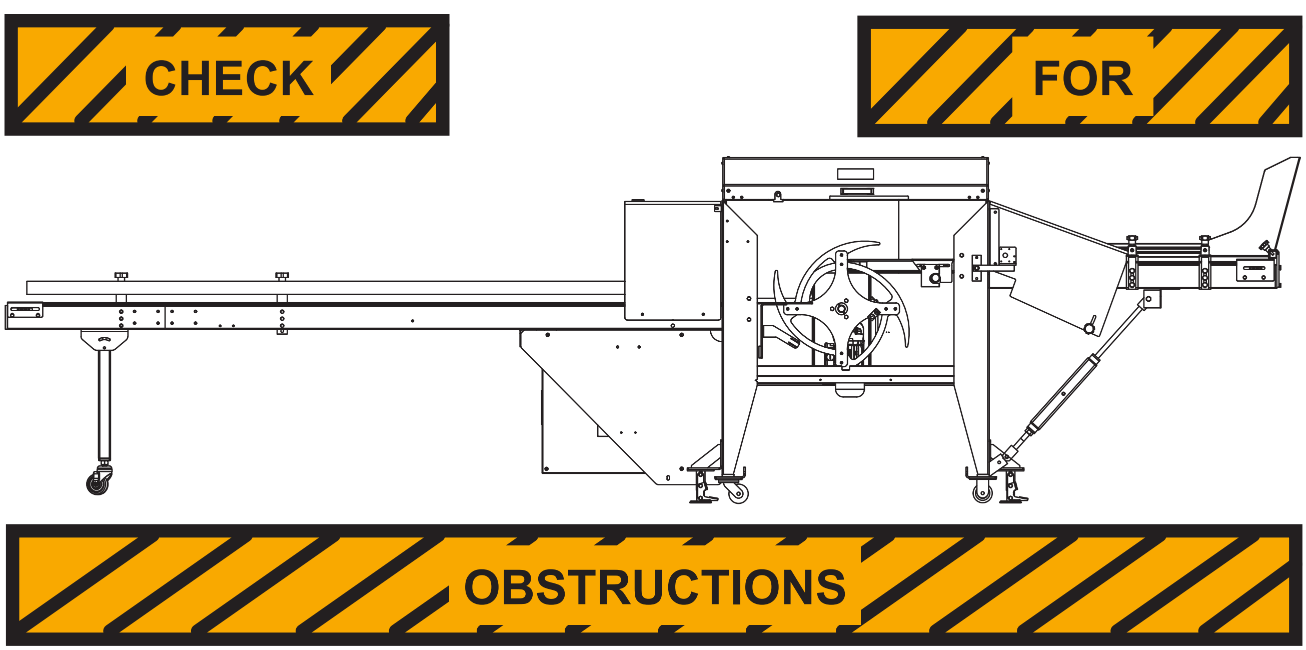
• All the emergency push-buttons and safety switches operate properly.

• All safety guards are in place.

• The main switch is engaged supplying power to the machine main enclosure.

• The machine is clear of obstructions, tools, trash, or other items.

• The machine product path is clear of disarrayed or out of specification product.

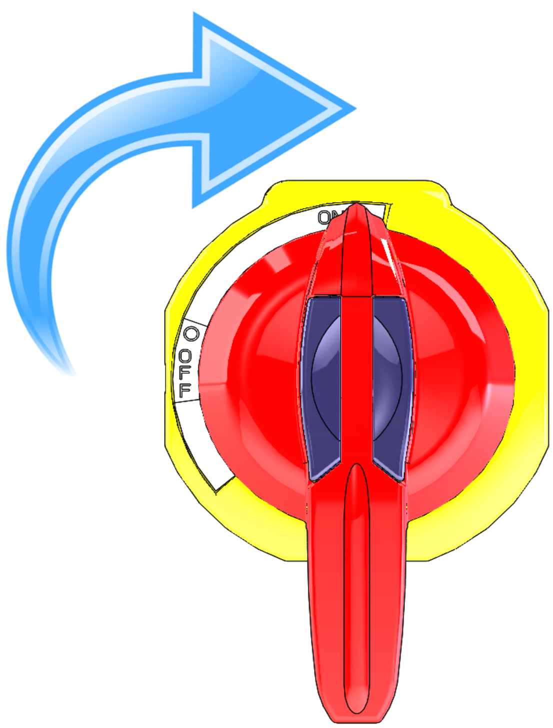


Start-up Procedure

Once all the pre start-up procedures are confirmed, the machine can be started as follows:

• Verify that the emergency stop push-button is pulled out.

• Unlock and put the electrical disconnect switch selector on the main cabinet into the ON position and then wait for the programmable logic controller (PLC) to initialize.



⚠ WARNING

Follow internal Lockout/Tag-out policy when Unlocking or locking the electrical disconnect switch.

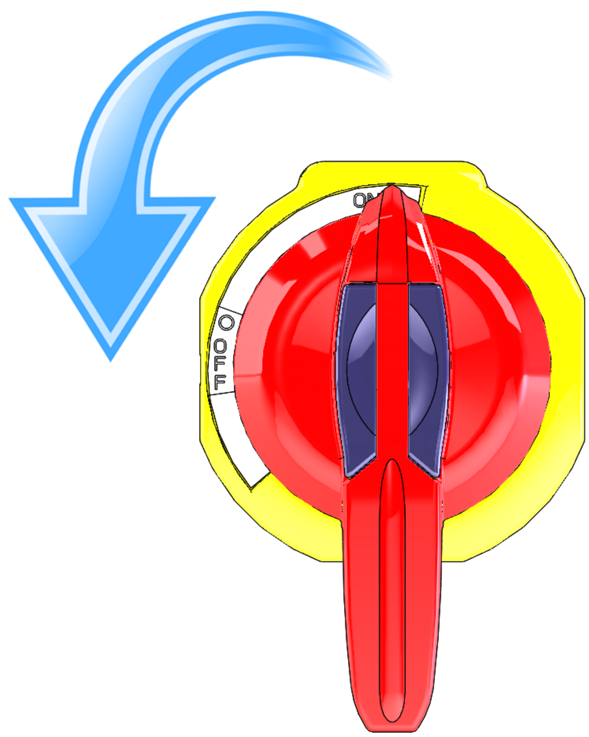
• Verify that the fault banner of the HMI is cleared of fault messages.

• Press the green On/Off button; the machine will start.

Shut-down Procedure

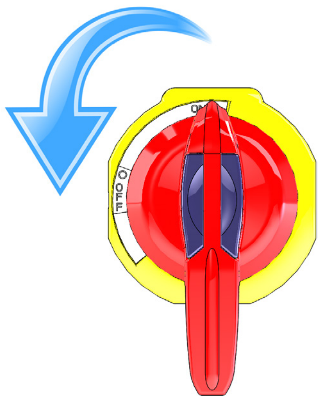
• Press the stop button (red while the machine is running).

• Switch the electrical disconnect to the off position and lock it.

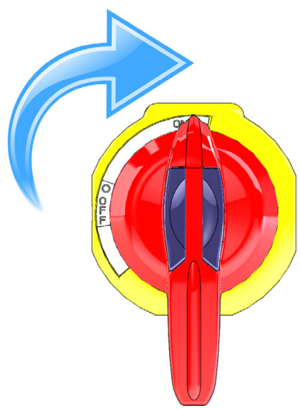


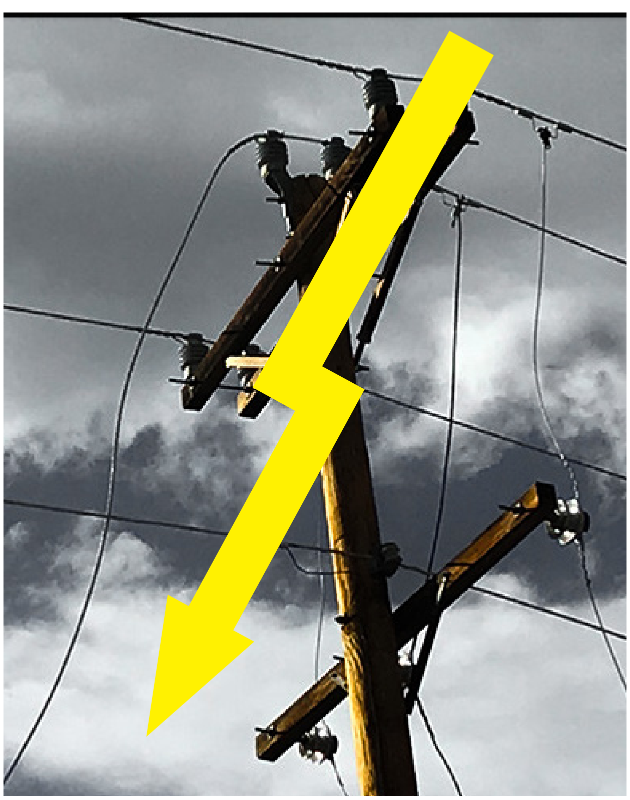
Recovering From A Power Outage

1. Turn off the machine for at least five minutes right after the power is restored. It doesn't matter how long the power outage lasted.



2. Turn it back on and follow the start-up procedures described above.





HMI Screens

Home Screen

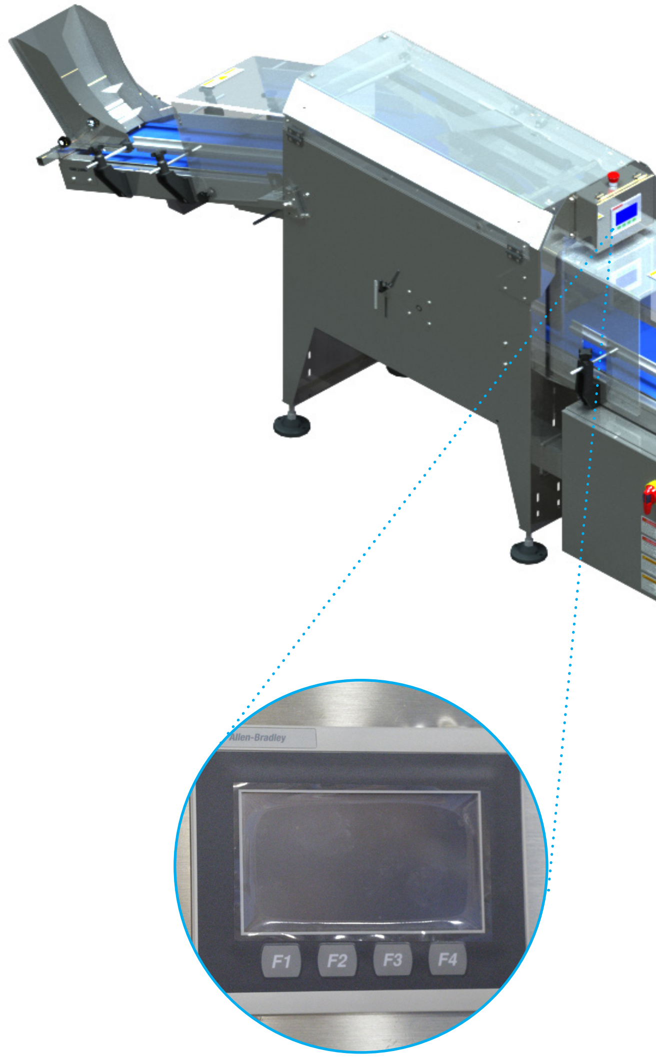
Power Push-button
Touching this icon turns on the control current allowing the operator to use the machine. Once the control current is on, the button will change to green; pressing it again will start the machine.

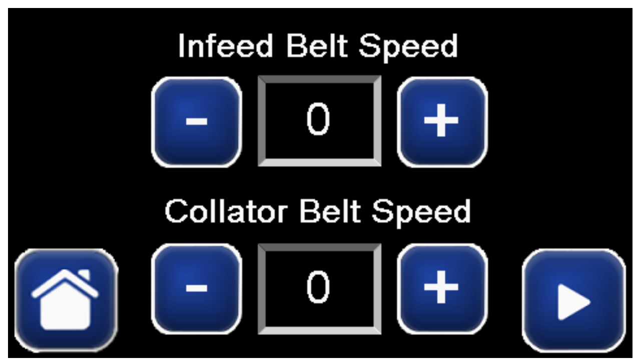
Information Icon
Touching this icon will take you to the Technical Data and Contact Information screens.

Data Storage Icon
Touching this icon will take you to the Production Data Screen

Warning Icon
Touching this icon will clear any displayed faults. Address the underlying problem which generated the fault(s) before clearing the fault(s).

Adjustment Icon
Touching this icon will take you to the Manual Adjustments Screen.





Belt Speed Manual Adjustments Screen

In-feed Belt Speed
Use blue ± buttons to manually adjust the in-feed belt speed, the current speed will be displayed in the centre button.

Collator Belt Speed
Use blue ± buttons to manually adjust the Collator (exit) belt speed, the current speed will be displayed in the centre button.

Home Icon
Touching this icon takes the operator back to the home screen.

Arrow
Touching this icon will take you to the next Manual Adjustments Screen.

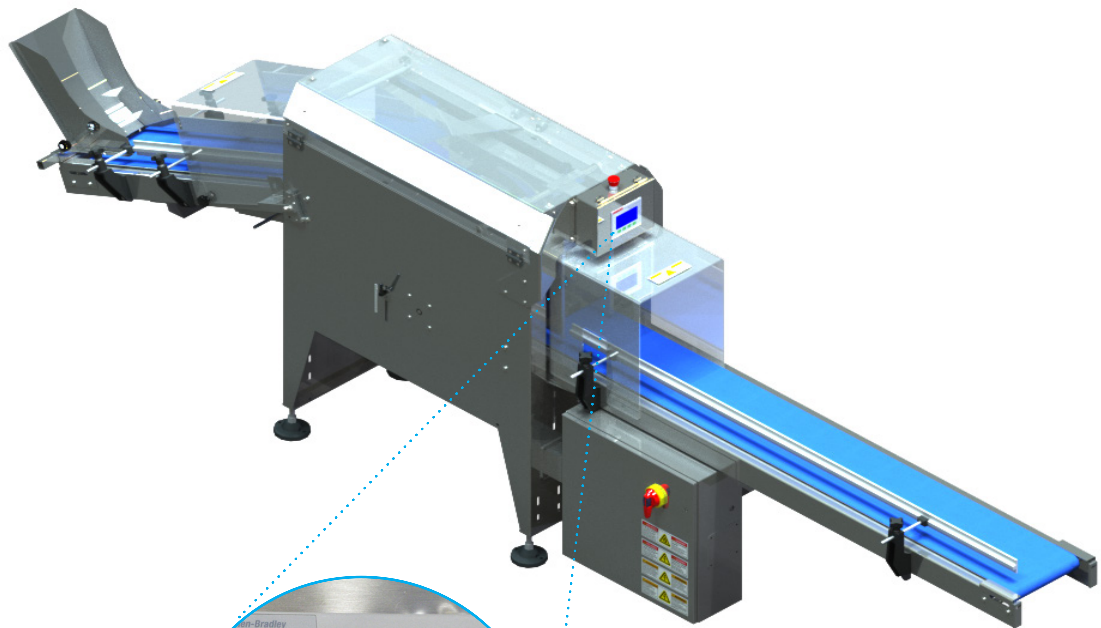
Collator Wheel Speed Adjustments Screen

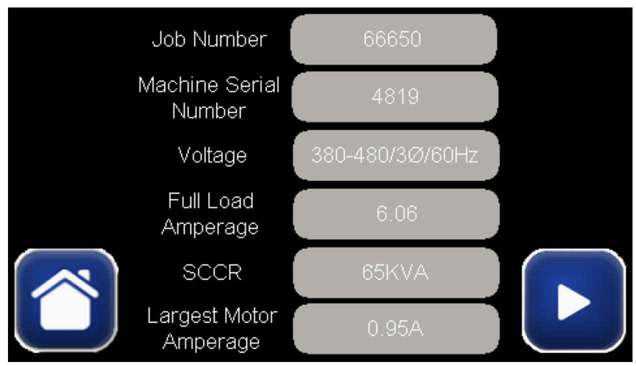
Collator Wheel Speed
Use blue ± buttons to manually adjust the collator wheel speed, the current speed will be displayed in the centre button.

Wheel Delay
Enter value in milliseconds (10-600) for the collator wheel delay.

Home Icon
Touching this icon takes the operator back to the home screen.

Arrow
Touching this icon will take you to the next Manual Adjustments Screen.





Machine Technical Information Screen

Job Number

The Job Number will be displayed here, it is a 5-6 digit numerical code. This number will be very helpful in locating the relevant files when making inquiries to **Blueprint** Automation regarding the machine.

Serial Number

This field will display a unique sequential number assigned to the machine by **Blueprint** Automation.

Voltage

This field will display the required voltage, phase, and frequency (**V** / **ø** / **Hz**) to be applied to the machine's electrical disconnect switch.

Full Load Amperage

This field will display the machine's current draw under full load in amperes (**A**).

SCCR

This field will display the short circuit current rating (**SCCR**) in kilovolt-amperes (**kVA**).

Largest Motor Amperage

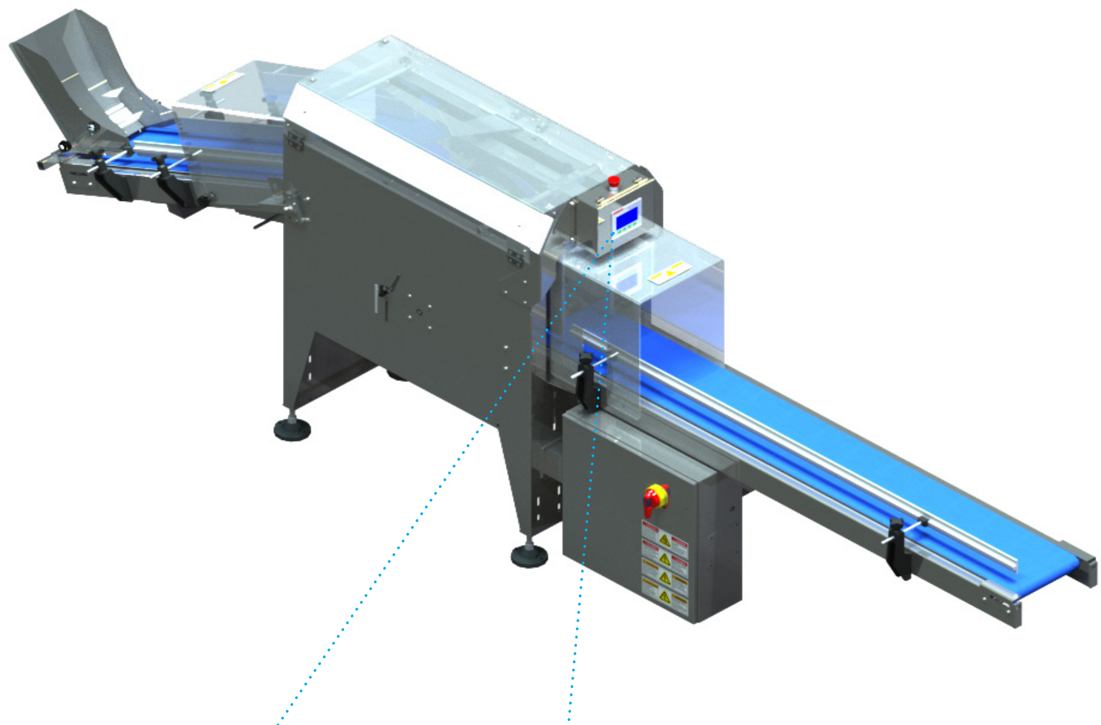
This field will display the current draw of the machine's largest motor in amperes (**A**).

Home Icon

Touching this icon takes the operator back to the home screen.

Arrow

Touching this icon will take you to the next Manual Adjustments Screen.



Machine Technical Information Screen

Contact Information

The first five fields will display **Blueprint** Automation's contact information.

Manufacture Date

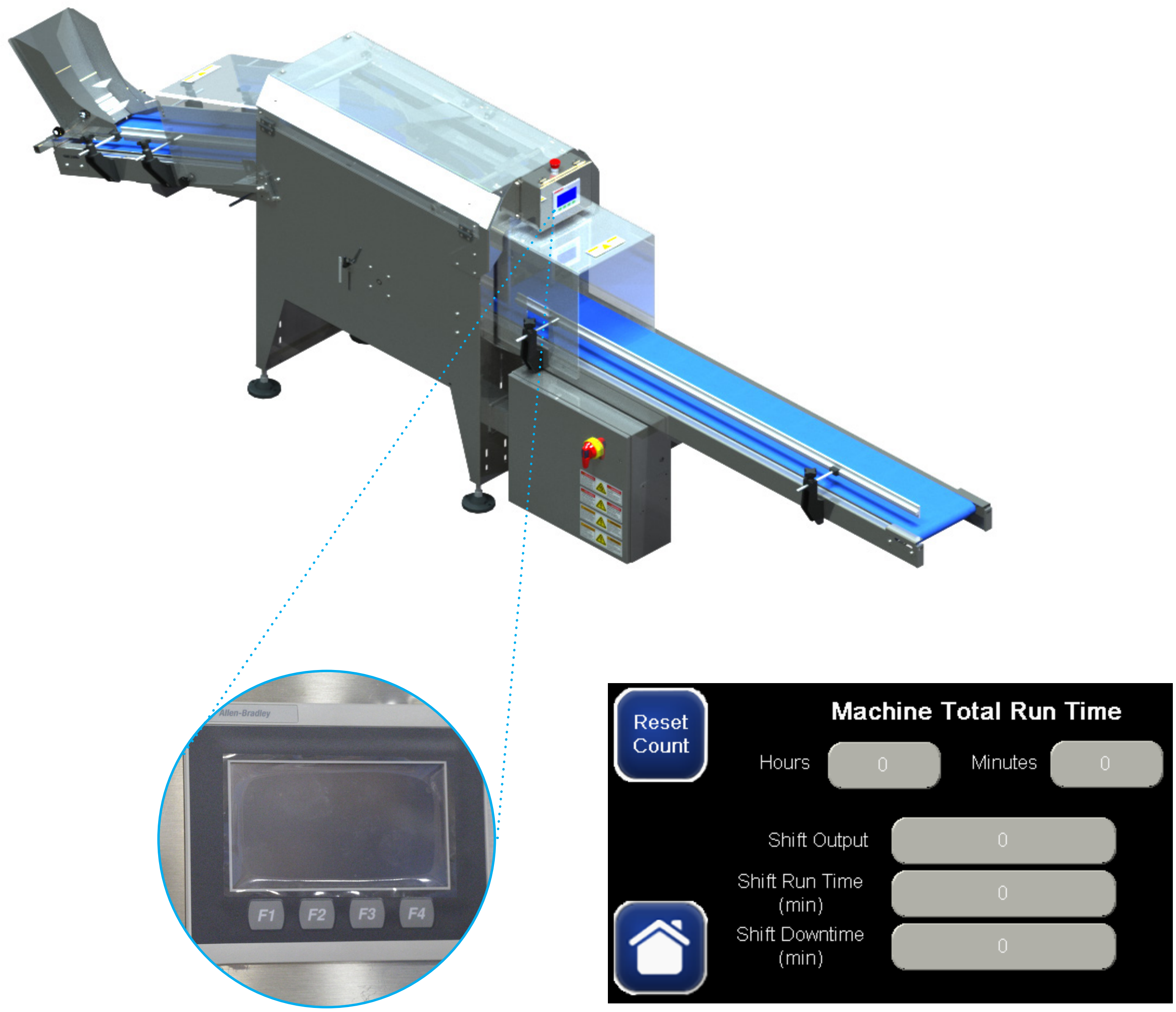
This field will display the date the machine was manufactured.

Home Icon

Touching this icon takes the operator back to the home screen.

Arrow

Touching this icon will take you to the next Manual Adjustments Screen.



Production Data Screen

Reset Output
Touching this icon will clear all the production data fields displayed on this screen.

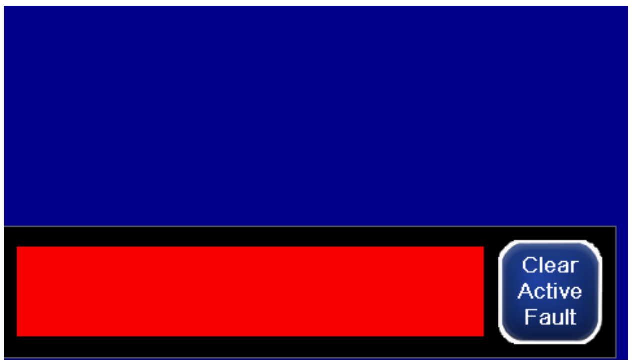
Production Data
The first seven fields on this screen will display he relevant production data accrued since the last reset.

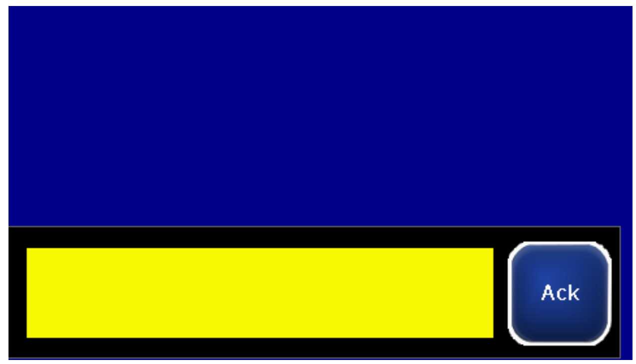
Home Icon
Touching this icon takes the operator back to the home screen.

Fault Messages

The following is a list the faults you may see on the HMI. When a fault is displayed, the yellow triangle on the left side of the screen will flash. After a fault is cleared the machine must be restarted at the Maine Start/Stop Screen.

Trigger	Value	Message	Display	Log
BagJamFault	1	Bag Jam Fault	Yes	Yes
CollateBeltVFDFault	1	Collate Belt VFD Fault. Check VFD0603.	Yes	Yes
CollateWheelVFDFault	1	Collate Wheel VFD Fault. Check VFD0513.	Yes	Yes
CollatorWheelFault	1	Collator Wheel Fault	Yes	Yes
Estop	1	Estop Pressed/ Door Open/Safety Not OK	Yes	Yes
InfeedVFDFault	1	In-feed Belt VFD Fault. Check VFD0503.	Yes	Yes
OverloadFault	1	Circuit Breaker Tripped	Yes	Yes





Bag Jam Fault

This fault will be displayed if the collator wheel becomes jammed with product packages. Press the emergency stop push-button and clear all packages from the collator wheel and in-feed belt. Reset the emergency stop push-button then press the green start touch cell on the home screen.

Collator Wheel Fault

This fault will be displayed if the collator wheel goes into a fault condition if the collator wheel photoelectric sensor moves out of adjustment or becomes defective.

Circuit Breaker Tripped

This fault will be displayed if a circuit breaker is tripped. Contact the maintenance department.

Emergency Stop Pressed

This fault will be displayed if you attempt to run the collator while the emergency stop push-button is activated. Reset the emergency stop push-button to enable running the system.

Safety Door Open

This fault will be displayed if a safety guard was left open.

VFD Fault

If this fault is displayed, it indicates a motor variable frequency drive (**VFD**) has entered a fault condition. Contact the maintenance department.

Operator Cleaning Procedures

Routine cleaning is important for optimal running conditions for the machine. Daily and weekly cleaning procedures to be undertaken by the operator are described below

At the Beginning of Each Shift

1. Using a low-pressure air line, blow any dust or debris off of the frame and conveyors. Carefully blow the dust off of the photocells and their reflectors.
2. Use a wet cloth - do not use alcohol or harsh detergents - to clean the conveyor belts off as they are jogged at reduced speed.
3. Use a dry cloth to wipe the face of the photoelectric sensors and their reflectors after using the air line on them.
4. After cleaning, cycle the machine to ensure proper operation.

NOTICE

Do not use alcohol or harsh cleaning agents as these may damage the machine or its components.

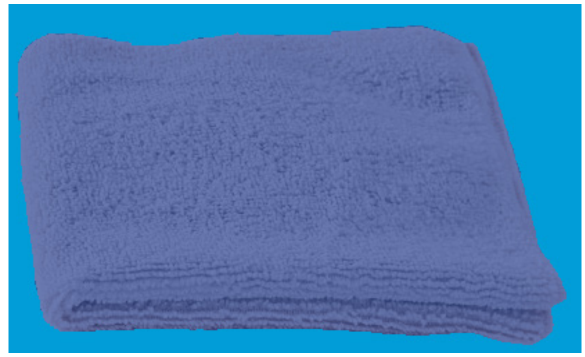
Weekly Cleaning

1. Using a low-pressure air line, blow any dust or debris off of the frame and conveyors. Carefully blow the dust off of the photocells and their reflectors.

⚠ CAUTION

Risk of eye injury from airbourne particulates, use protective eye-wear when working with compressed air jets.

2. Use a wet cloth to carefully clean the conveyor belts off as they are jogged at reduced speed. If the belts are especially dirty, use a light soapy mixture to wipe down the belts.



3. With the machine locked out, using a bucket of slightly soapy water and cleaning cloth, wipe down the framework of the machines. Lift the conveyor belts and clean the pans underneath them. Wipe down all surfaces with product build-up on them. Afterwards, use the air hose to lightly blow off the water to dry the surfaces. Then use a dry cloth to finish up.
4. When cleaning the floor under the equipment, be careful not to splash the pressurised water towards the equipment. This can result in water being forced into drainage holes or onto flat surfaces and harsh cleaning solutions getting into the equipment.

NOTICE

DO NOT direct the pressurised water at motors or bearings as the seals can be compromised and water will get inside. If a clear rinse is necessary, use non-pressurised water in a downward flow and dry off afterwards.